Overview

Welcome to Week 7! So far, our models have focused on static, independent data like images. But how do we handle data that unfolds over time, such as sentences, speech, or video?

This week, we are diving into sequence modeling. We will explore architectures designed to maintain a "memory" of past inputs to understand context, and see how they power modern natural language processing and multimodal systems.

Here is what we will cover this week:

- RNNs and LSTMs: How unfolding computational graphs allow networks to process sequential data, and how Long-Short Term Memory (LSTM) solves the vanishing gradient challenge of long-term dependencies.
- Word Embeddings & Text Classification: How we translate words into dense mathematical vectors to teach machines to understand context and sentiment.
- Multimodal Applications: Applying these sequence models to real-world tasks like speech recognition and combining them with our vision architectures for image captioning.

Learning Outcomes

- Deconstruct the mechanics of recurrent architectures (RNNs and LSTMs) to explain how they maintain temporal memory and overcome the vanishing gradient problem when processing sequential data.
- Formulate sequence-to-sequence solutions utilizing word embeddings and attention mechanisms to tackle complex natural